ARYABHAT FOUNDATION · CURRICULUM LAB

Sky & Space Activity Handbook

Every hands-on activity in the Class 6–9 astronomy courses, in teaching order — with the concept each one teaches and the kit it needs.

107 activities **45** sessions with hands-on work **4** classes

An activity is listed in full the FIRST time its concept is taught; when a later class returns to it, the handbook points back rather than repeating. Companion booklets: one per course, in files/2026/courses/.

CLASS 6

Sky & Space: Class 6

A year-long module for class 6 in regular school periods: from a spinning ball underfoot to India in space, built on shadow sticks, a moon diary, and one sky night.

Kit for the year: Projector · Paper, pencils, protractor · Gnomon / shadow stick · Measuring tape · Star chart / planisphere · Red torch

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1 A ball, spinning

Classroom · 40 min

Bring: Projector, Paper, pencils, protractor

Globe treasure hunt: find India, both poles, the equator, and the tilt

Reading the Globe — The globe is Earth's honest model: a tilted ball with poles, continents and oceans. Finding India, the poles and the tilt on a globe comes before any talk of rotation or seasons.

Spin tops and a bicycle wheel: find each one's axis; watch a tilted top wobble

Spin & Axis — A spinning top turns around an invisible line — its axis — and the axis itself stays steady while everything else whirls. Earth is a giant top; its axis points at the pole star.

Globe + torch: spin the globe, watch a sticker city pass through day and night

Day & Night — Earth spins once in about 24 hours, so the Sun appears to rise in the east and set in the west. Day and night are a matter of which side of Earth faces the Sun.

> Darken the room, one torch, one globe, and put a sticker on your city — day and night must happen to THEM, not to a diagram. Expect the classic wrong answer that the Sun goes around us or that it 'switches off'; let a student rotate the globe and discover their sticker carrying them into shadow. Do not mention orbits, seasons or tilt-consequences yet — one revolution of the globe is the whole lesson, and it is enough.

2 The dome overhead and the Sun's road

Indoor lab · 80 min

Bring: Paper, pencils, protractor, Gnomon / shadow stick

Umbrella planetarium: draw stars inside an umbrella and rotate it

The Sky Dome — The sky looks like a dome overhead: the horizon is its rim, the zenith its top. Everything we observe is a direction on this dome, not a distance.

Shadow-stick compass: mark shadow tips through a day

Cardinal Directions — Finding north, south, east and west from the Sun and shadows. Sunrise and sunset directions drift through the year — due east/west only near the equinoxes.

Chart shadow length every hour on the school ground

Sun's Daily Arc — The Sun climbs from the eastern horizon, peaks toward the south (from India), and sinks west. Shadow lengths and directions trace this arc.

> Take the double period so you can step into the school ground twice, an hour apart, and mark the Sun's position against a fixed landmark — the arc has to be SEEN moving, not asserted. Fix the cardinal directions to the school building itself ('the gate is west') so every later session has a shared compass. Most children believe the Sun rises exactly east every day; plant the doubt now, the seasons session will harvest it.

3 A stick tells the time

Indoor lab · 40 min

Bring: Gnomon / shadow stick, Measuring tape, Paper, pencils, protractor

Torch + toys: predict each shadow's shape and direction before switching on

Light & Shadow — Light travels in straight lines, so an object blocks it and casts a shadow shaped like itself. Shadow direction and length depend on where the light is — the key to sundials and eclipses.

Find local noon with a plumb stick; compare with clock noon

Gnomon & Local Noon — A vertical stick tells time: its shadow is shortest at local noon, pointing exactly north–south. The gnomon (shanku) is astronomy's oldest instrument.

Calibrate your hand: fist ≈ 10°, spread hand ≈ 20°

Angles on the Sky — Sky positions and sizes are measured in angles: your fist at arm's length spans about 10°, the Moon half a degree. Degrees, arcminutes and arcseconds are the astronomer's ruler.

> Plant the gnomon before school and have monitors mark the shadow tip every period — by this class there is a curve on the ground and the shortest shadow points true north. Teach fist-at-arm's-length = 10° here, on the gnomon's shadow angle; the class will use it all year. The stick stays planted: it is the course's only permanent instrument, and sessions 4 and 6 come back to it.

4 How big is the Earth?

Indoor lab · 80 min

Bring: Gnomon / shadow stick, Measuring tape, Paper, pencils, protractor, Projector

Draw the classroom to scale; then the school grounds

Maps & Scale Models — A map shrinks the real world by a fixed scale — 1 cm standing for 1 km — keeping shapes and directions honest. Star charts, globes and solar-system walks are all the same idea.

Locate five cities from coordinates alone; read your town's off a map app

Latitude & Longitude — An address grid for the whole planet: latitude counts degrees from the equator, longitude from Greenwich. Bhopal sits near 23°N, 77°E — and your latitude is written in your sky.

List and debate the classical evidences

Earth is a Sphere — Ships sinking hull-first, new stars rising as you travel south, and Earth's always-round shadow on the eclipsed Moon — classical proofs, no spacecraft needed.

Partner-school Eratosthenes measurement at local noon

Measuring the Earth — Eratosthenes sized the Earth around 240 BCE with two sticks and a noon shadow angle. Two schools at different latitudes can repeat it today over WhatsApp.

> Start from the school map they drew in geography and shrink outwards — city, country, globe — so scale is a habit before it is a number. Then Eratosthenes: same stick, same noon, a partner school a few hundred km north or south, and the class computes the circumference of the planet themselves. Arrange the partner school weeks in advance; a WhatsApp photo of their noon shadow is all you need. If the numbers come out within 15% the room goes silent — do not fill the silence.

5 Why summer? (Not what you think)

Classroom · 40 min

Bring: Projector, Paper, pencils, protractor

Torch at a slant vs straight-on: same light, bigger patch

Seasons — Earth's axis is tilted 23.4°, so through the year each hemisphere leans toward or away from the Sun — changing day length and the Sun's height. Distance from the Sun is not the cause.

Mark sunrise direction on the horizon each month

Equinoxes & Solstices — Four turning points of the year: two equinoxes when day equals night, two solstices when the Sun's arc is highest or lowest. They anchor seasons and calendars worldwide.

> Before teaching anything, have every child write one sentence explaining summer. Most will say the Earth gets closer to the Sun — keep the slips. Globe and lamp, tilt fixed, walk the orbit: let a student notice that 'closer' would make BOTH hemispheres hot at once, while Australia's December beaches say otherwise. Return the slips at the end and let them mark their own. The gnomon's noon shadow, now weeks long in your logbook, shows the solstice drift for free.

6 The noon shadow project

Solar session · 40 min

Bring: Gnomon / shadow stick, Measuring tape, Paper, pencils, protractor

Trace the Tropic of Cancer across an India map — which towns does it touch?

Tropics & Polar Circles — The Tropic of Cancer — which crosses India through MP and Gujarat — marks where the noon Sun can stand exactly overhead. These lines are the axial tilt drawn onto the map.

Find your town's zero-shadow dates; photograph a bottle at true noon

Zero Shadow Day — Twice a year, everywhere between the tropics, the noon Sun stands exactly overhead and shadows vanish for a moment. India's favourite mass astronomy experiment — date depends on your latitude.

> A daytime outing to the faithful stick: measure the noon shadow carefully and compute your latitude's distance from the Tropic of Cancer. If your school sits south of the Tropic, find your actual zero-shadow date from the tables and move THIS session to match it — a vertical stick with no shadow is the single most photographed moment of the year. North of the Tropic, teach it as the shadow that never quite dies, and why that line on the map is drawn where it is. Never let anyone stare at the Sun; the shadow is the instrument.

7 Start your moon diary

Classroom · 40 min

Bring: Paper, pencils, protractor

Moon diary: sketch the Moon and note time/direction for 30 days

Observing the Moon — The Moon is visible in the day as often as at night, rises about 50 minutes later each day, and shows changing shapes. Just watching it for a month is a complete experiment.

> One concept, one period, and a month of homework: every child sketches the Moon each clear night — shape, time, and which side is lit — in a printed 30-box diary. Teach them that the Moon is up in the DAYTIME half the month, which most adults do not know, so morning sketches count. This session exists to start the clock: the phases lesson is deliberately four weeks away, because the Moon is the only teacher of its own rhythm. Chase the diaries weekly or the lab on day 87 falls flat.

8 Phases, from inside the orbit

Indoor lab · 80 min

Bring: Paper, pencils, protractor

Lamp + ball around your head: see all phases in one minute

Phases of the Moon — The Moon is always half-lit by the Sun; the phase is how much of that lit half faces us, cycling in about 29.5 days. Phase, rise time and Sun–Moon angle all lock together.

Scale model: football Earth, tennis-ball Moon 7 m apart

Moon Orbits Earth — The Moon circles Earth roughly monthly at about 384,000 km — around 30 Earth-diameters away. Its orbital motion is what drives the phase cycle and its nightly 13° eastward drift.

For one model (globe/orrery), list: what it gets right, what it gets wrong

Models in Science — A globe, a map, a diagram of the Sun — none are the real thing; all are models that answer some questions well and others not at all. Science improves by testing models against observation.

> Darken the hall: one lamp is the Sun, every child holds a ball at arm's length and turns on the spot — each of them is the Earth watching their own moon wax and crescent. Phases must be seen from inside the orbit; a diagram drawn from above teaches the teacher, not the child. Then lay their month of diaries end to end and match diary to ball. The misconception to kill on sight: 'phases are the Earth's shadow'. Do not kill it with words — ask where the shadow would have to be, and let the lamp answer. Close by naming what you just did: the lamp is not the Sun and the ball is not the Moon, yet they answered a real question — that is what a scientific model IS, and this class will use models all year.

9 The sky night

Observing night · 80 min

Bring: Star chart / planisphere, Red torch

Colour hunt: find one blue, one yellow, one red star

Stars Differ — Stars are not identical dots: they differ in brightness and in colour — steely blue Vega, golden Arcturus, red Betelgeuse. These differences are real physics, visible to the naked eye.

Watch Orion or the Big Dipper shift over 2 hours

Diurnal Motion — The whole starry sky wheels westward through the night, circling a fixed point — the celestial pole. This is Earth's rotation seen from the inside.

Measure Polaris altitude with a fist-and-thumb protractor

The Pole Star — Polaris (Dhruva) sits nearly at the north celestial pole, so it barely moves and always marks north. Its altitude above the horizon equals your latitude.

> The one evening the school will give you — spend it on the naked eye, not on a telescope queue. A crescent moon greets them early (point out the earthshine even though you are not formally teaching it) and sets before it can wash out the constellations. Anchor two constellations, then return to them after an hour so the class SEES the sky wheel; only then reveal the one star that did not move. Expect disbelief that Dhruva is faint — that surprise is the lesson. Give the Indian names alongside the Greek ones as equals, not as trivia. Have a cloud date agreed with the school in advance.

10 When shadows swallow the Moon and Sun

Indoor lab · 80 min

Bring: Projector, Paper, pencils, protractor

Blackout-box experiment: can you see the object with zero light?

How We See — We see things only when light from them enters our eyes — the eye receives light, it doesn't send anything out. No light, no seeing; more light gathered, fainter things seen.

Pinhole projection of the Sun

Solar Eclipses — At new moon, the Moon's small shadow can sweep across Earth, briefly hiding the Sun. Totality happens because the Moon and Sun span the same half-degree — a cosmic coincidence.

> Same lamp and balls as the phases lab, deliberately — an eclipse is what happens on the rare month the three actually line up, and the class already owns the model that shows why it is rare. Contrast against phases explicitly: last month's wrong answer ('Earth's shadow') is this month's right one, and saying so out loud inoculates both. Teach eye safety with full weight here: a solar eclipse is the one astronomical event that can injure a child, and the pinhole projector they build in the last twenty minutes is their licence to watch one. Address the eclipse-food superstitions directly and without mockery — the children will hear them at home.

11 The wanderers and the family

Indoor lab · 80 min

Bring: Projector, Paper, pencils, protractor, Measuring tape

Plot a planet's position against stars weekly for a month

Planets: the Wanderers — A few bright 'stars' drift against the fixed constellations from week to week — the planets. They shine steadily while true stars twinkle.

Powers-of-ten zoom: classroom → city → Earth → Sun in ×10 jumps

Big Numbers & Powers of Ten — A lakh, a crore, a billion, 10^8 — big numbers need both names and notation. Astronomy runs on powers of ten; without them, its distances are just meaningless words.

Peppercorn solar-system walk across the school ground

Scale of the Solar System — With Earth as a peppercorn, the Sun is a football 25 m away and Neptune is 750 m down the road. The emptiness is the real lesson; the AU is the ruler.

> Start from the sky night: everything they saw stayed fixed in its constellation — except, had they watched for weeks, a few bright 'stars' that drift. That drift, not a poster of eight balls, is what a planet IS, and it is why five of them have been known since before history. Then take the family tour, but keep insisting on the difference between what we see (dots that wander) and what we know (worlds). Walk the planet order down the corridor first — sequence only. Then do it honestly: Earth a peppercorn, the Sun a football 25 m away, Neptune 750 m down the road. Every textbook poster lies about this, and the walk is the one lesson about the solar system nobody forgets: it is made of emptiness.

12 India in space

Review & assess · 80 min

Bring: Projector, Paper, pencils, protractor

Mission timeline wall: plot every ISRO science mission

India in Space — Chandrayaan-3 landed at the lunar south pole, Mangalyaan orbited Mars on a first attempt, Aditya-L1 stares at the Sun, AstroSat scans the X-ray sky — and Gaganyaan comes next.

Read and unpack two verses of the Aryabhatiya

Aryabhata's Astronomy — In 499 CE Aryabhata taught that Earth rotates, gave π to four decimals, built sine tables, and explained eclipses by shadows — not demons. The namesake of this foundation.

> The closing session joins the two ends of the year: Aryabhata computed the Earth's spin and explained eclipses by shadows fifteen centuries before the satellite named after him flew — and the class has personally redone both of those pieces of work with a stick and a lamp. Run it as review-by-storytelling: Chandrayaan needs their moon diary, Aditya needs their eye-safety rules, Mangalyaan needs their wanderers. End by having each child present one measurement THEY made this year. The message to land: this subject is not finished, and it is theirs.

CLASS 7

Sky & Space: Class 7

The second year for classes that did Sky & Space 6: light, the Sun as an engine, tides and craters, and the year the class earns a telescope — ending at the eyepiece.

Kit for the year: Projector · Paper, pencils, protractor · Binoculars · Dobsonian reflector · Eyepiece set · Solar filter (front aperture) · Measuring tape · Star chart / planisphere · Red torch · Smartphone · Phone-to-eyepiece adapter · Tripod · Gnomon / shadow stick

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1 Light is a traveller

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor

Count seconds between lightning and thunder; compute the storm's distance

Speed, Distance & Time — Speed = distance ÷ time: know two, find the third. Thunder after lightning gives the storm's distance — the same trick, scaled up, turns light's travel time into cosmic distances.

Light-delay role play: 'messages' walking across a scale solar system

Light & Lookback Time — Light is fast but finite — 8 minutes from the Sun, 4 years from the nearest star. Every telescope is a time machine: we see objects as they were, not as they are.

Laser + water tank refraction demo

Reflection & Refraction — Mirrors bounce light by a fixed rule; glass bends it as it slows down. These two behaviours are the entire toolkit from which every telescope is built.

Make a rainbow with a water tray, mirror and sunlight

Rainbows & Colours — White sunlight is a mixture of all colours — raindrops or a prism fan it out into a rainbow. Colour is a property of the light itself, not paint added to it.

> Open with the eight minutes: the sunshine on their desk left the Sun before the period started. Speed-distance-time is maths class made real — have them compute the Moon's 1.3 seconds themselves before you reveal it. Then a mirror, a glass of water with a pencil in it, and a prism at the window: bounce, bend, split. Do not name lenses or telescopes yet, but know that this entire year rests on this one period — every instrument they will touch is these three tricks in a tube.

2 The Sun is an engine

Classroom · 40 min

Bring: Projector, Paper, pencils, protractor

Trace your lunch's energy back to the Sun, step by step

The Sun Powers Earth — Sunlight drives nearly everything here: plants, winds, rain, and the warmth of the day. Grasping the Sun as Earth's engine is the first reason to care where it is in the sky.

Lux-meter (phone app) vs distance from a bulb: plot $1/d^2$

The Inverse-Square Law — Double the distance, quarter the light — brightness falls as $1/d^2$. This single rule connects a star's true power, its distance, and how bright it looks to us.

> Every joule in the room — the food, the fan, the child — is rerouted sunshine; trace one chain (sun → grass → roti → cricket shot) on the board and let them build two more. Then inverse-square with a torch and a grid drawn on card: double the distance, a quarter the brightness, measured not preached. Land the punchline on the solar system they already know: this is WHY Venus bakes and Neptune freezes, and they can now say by how much.

3 Glass: lenses, mirrors, and the telescope

Indoor lab · 80 min

Bring: Binoculars, Dobsonian reflector, Eyepiece set, Paper, pencils, protractor

Project the Sun or a window with a magnifier; measure focal length

Lenses, Mirrors & Focal Length — A curved lens or mirror gathers parallel starlight to a focus; the focal length sets image scale. Real images on a screen — this is where 'optics' becomes 'instrument'.

Strip-down demo of a Newtonian: follow the light path

Refractors & Reflectors — Refractors use a lens, reflectors a mirror, catadioptrics both. Each design trades cost, size, and image quality differently — there is no single 'best telescope'.

Binocular tour: Moon, Pleiades, a Milky Way sweep

Binocular Astronomy — A 7×50 binocular — 7× magnification, 50 mm objectives — shows Moon craters, Jupiter's moons and hundreds of clusters. The best first instrument, and many observers' favourite forever.

Finder-alignment and focus drill on a distant treetop, then the Moon

First Light: Using a Telescope — Align the finder in daylight, start with the lowest power, centre on something bright, focus slowly. Ten minutes of drill separates a magical first night from a frustrating one.

> A magnifier focusing the window onto a sheet of paper — a real, upside-down, full-colour image of the world on a page — is the moment optics stops being a diagram. Build from it: two lenses make a refractor, a curved mirror makes their Dobsonian. Then hands on the actual instruments, in daylight, at a distant building: finding, focusing, and the discipline of not touching glass. Every child moves the Dobsonian and focuses the binoculars TODAY; the sky night is not the place to learn where the focuser is. Say out loud that magnification is the least important number — they will meet the 500x marketing lie at home.

4 Our star, up close and safely

Solar session · 80 min

Bring: Dobsonian reflector, Solar filter (front aperture), Paper, pencils, protractor

Scale comparison: Sun vs Betelgeuse vs Earth

The Sun is a Star — Our Sun is an ordinary star seen close-up — a million-km fusion furnace. Every night-sky star is a sun; many have their own planets.

Track a sunspot group across the disc for a week

Sunspots & Solar Activity — The Sun's magnetic weather: spots, flares and mass ejections rising and falling on an 11-year cycle. It drives aurorae and can rattle satellites and power grids.

Eyepiece-projection sunspot count

Solar Observing (Safely) — NEVER look at the Sun without certified filters — projection and proper solar film show sunspots safely; hydrogen-alpha scopes reveal prominences. Safety protocol is the first lesson.

> The Sun is a star; the stars are suns — say it once at the start and once at the end, because it reorders the whole sky in their heads. Projection first: the whole class around one projected disc, hunting sunspots, before anyone queues at the filtered eyepiece. Safety is theatre and must be theatrical: inspect the front-aperture filter in front of them, cap the finder in front of them, and tell them about the eyepiece-filter cracks of old. Sunspots bigger than the Earth, counted by children — and if the disc is blank that day, teach them that the quiet Sun is a real observation too. Log the count; the aurora session will want it.

5 Lights the atmosphere makes

Classroom · 40 min

Bring: Projector, Paper, pencils, protractor

Height ladder: kite, cloud, plane, ISS, Moon — order them by distance

Weather vs the Sky — Clouds, rain and haze live a few kilometres up; the Moon, planets and stars lie far beyond. Weather hides the sky but never touches it — astronomy begins where weather ends.

Scale drawing: on a 30 cm Earth, how thick is the atmosphere? (a pencil line)

The Ocean of Air — We live at the bottom of a thin skin of air that thins with height and fades into space around 100 km up. It burns meteors, blurs stars, bends light — and keeps us alive.

Halo watch: verify the ring's radius with an outstretched hand

Halos & Coronae — A big 22° ring around the Moon or Sun is a halo from ice crystals; a small rainbow-fringed disc hugging the Moon is a corona from water droplets. Free atmospheric optics, often before rain.

Track a solar storm's Kp index and aurora alerts in real time

Aurorae — Solar-wind particles funnelled down Earth's magnetic field make the upper air glow in curtains of green and red. Strong storms have pushed aurorae as far south as Ladakh.

> The sorting lesson: clouds, rain and rainbows live in the bottom few kilometres; the Moon and stars live behind all of it — most children file 'sky' as one drawer and this period gives them two. Halos are the payoff: a ring round the Moon is ice crystals in OUR air, not something the Moon does, and once told, children start reporting them within the fortnight — put a halo-report corner on the classroom wall. Close with aurora as the two drawers connecting: the Sun's tantrums (their own sunspot log) lighting the top of the air. Be honest that they will likely never see one from India; honest scarcity beats false promise.

6 The Moon pulls, the Moon remembers

Indoor lab · 80 min

Bring: Projector, Paper, pencils, protractor, Measuring tape

Crater experiment: drop stones into flour + cocoa at different speeds

Craters & Maria — The Moon's dark 'seas' are ancient lava plains; its craters are asteroid scars, preserved for billions of years with no air or water to erase them. The face of the Moon is a history book.

Case study: Lonar crater — map, rock evidence, age

Impacts on Earth — Earth gets hit too — Lonar Lake in Maharashtra is a 50,000-year-old impact crater, and the dinosaurs met a 10-km asteroid. Erosion, not luck, hides our scars.

> Two faces of the same neighbour. First the pull: the Moon's gravity stretches the ocean, and the trap to disarm is 'one bulge' — there are two, and a child holding the diagram will ask why; have the answer ready (the far side is left behind, not pushed). Then the scars: flour tray, cocoa dusting, and marbles dropped from different heights — every crater is round regardless of impact angle, which is the question someone always asks. Their craters against Chandrayaan photographs, then the sting: the Moon keeps every scar for four billion years because it has none of the air and weather we sorted out in the last session. Earth got hit just as hard and healed.

8 Learning to see in the dark

Indoor lab · 80 min

Bring: Star chart / planisphere, Red torch, Paper, pencils, protractor

Colour-vanishing test: watch colours die as light dims

The Eye as a Detector — Cone cells see colour but need light; rod cells see faint things only in grey. That's why nebulae look ghostly-grey in the eyepiece yet glow red and blue in photographs.

Red-torch discipline on a star party; averted-vision test on a faint cluster

Dark Adaptation & Averted Vision — Eyes take 20–30 minutes to reach full night sensitivity, and one white-light glance resets them. Looking slightly beside a faint object uses your more sensitive off-centre retina.

Make a paper planisphere

Star Charts & Planetarium Apps — Reading a sky map: orientation, scale, magnitudes, and matching chart to sky. A planisphere or app turns the catalogue sky into 'what's above me right now'.

Design a club log sheet; log every session for a month

Observing Logs & Sketching — Date, time, sky conditions, instrument, and what you actually saw — a log turns stray sessions into a record of growth, and sketching trains the eye to see more.

> The unglamorous session that decides whether the sky night succeeds. Blackout the lab and spend real minutes in the dark: let them experience faint things surfacing as their eyes adapt, then ruin it with one white torch flash — now the red torch needs no policing, they have felt why. Planisphere drill as a game: set the date and hour, race to name what is up tonight. Star names give the chart its handles; the observing log is the year's closer — rule the first page together and make the sky night its first entry. A log they keep is the difference between an outing and an astronomer.

9 Night at the eyepiece

Observing night · 80 min

Bring: Dobsonian reflector, Binoculars, Eyepiece set, Star chart / planisphere, Red torch, Smartphone, Phone-to-eyepiece adapter, Tripod

Sketch one crater on three nights as shadows move

Lunar Observing — Craters, maria, mountain shadows — the Moon is best along the terminator, where low sunlight throws relief into drama. Full moon is actually the flattest, worst time.

Spot and photograph earthshine on a 2–3 day old crescent

Earthshine — The ghostly glow filling the dark part of a crescent Moon is sunlight reflected off Earth — 'the old Moon in the new Moon's arms'. You are seeing our planet light another world.

Phone-at-eyepiece Moon shot on a club night

Smartphone Astrophotography — A phone held (or bracketed) at the eyepiece captures the Moon and planets; night modes catch constellations and even the Milky Way. The zero-cost gateway to imaging.

Dark-sky trip: map the band against constellations

The Milky Way Band — From a dark site a faint luminous band arches across the sky — the combined light of billions of stars in our galaxy's disc, seen from inside it.

> The year's summit, and the crescent is chosen, not luck: a terminator full of shadowed craters early, earthshine to explain while they queue ('the ghost light is Earth shining on the Moon's night'), and it sets in time to leave the sky dark for the faint stuff. Run the Moon at the Dobsonian first and let every child take one phone photo through the eyepiece — the adapter beats hand-holding, and that photo does more for the subject at home than any circular. Then binoculars up the Milky Way band to a bright cluster: teach them it is grainy with stars, a thing no screen conveys. Everyone writes their log entry ON SITE, red torch, before the bus — memory edits, logs do not. Cloud date agreed with the school in advance.

10 Build a sundial, own the week

Hands-on / build · 80 min

Bring: Gnomon / shadow stick, Paper, pencils, protractor, Measuring tape, Projector

Build a cardboard equatorial sundial for your latitude

Building Sundials — A sundial's gnomon must parallel Earth's axis — tilted at your latitude, aimed at the pole. Get that right and horizontal, vertical or equatorial dials all read true.

Derive the weekday order from the planetary-hours rule

Vara & the 7-Day Week — The week comes from the seven classical 'planets' ruling hours in rotation — Ravivar/Sunday to Shanivar/Saturday match across India, Rome and Babylon. Astronomy fossilised in every calendar.

Match each graha to what it really is; spot all seven visible ones in a year

The Navagrahas — The nine 'grahas': Sun, Moon, five visible planets, plus Rahu and Ketu — not nine planets, but the nine movers of the sky as ancient India catalogued them.

> The closing workshop ties this year's optics-and-instruments spirit back to class 6's shadow stick: the gnomon must now be TILTED to the pole star's height — their latitude, which they measured last year — and suddenly one dial-plate works all year. Each pair builds and calibrates a card sundial against the school's true-north line at noon. Then the aha that hides in the calendar: Sunday, Monday, Saturday — the week is the seven moving lights of the navagraha, sitting in their pocket diary all along. Teach Rahu and Ketu as the eclipse points, the old astronomers' bookkeeping for a real geometry. Present it as heritage of observation; the astrology argument gets its own airing in a later year, when they can follow it.

CLASS 8

Sky & Space: Class 8

A year of naked-eye astronomy for class 8: read the night sky with charts and handspans, understand light and the instruments that catch it, then follow that light out past the Sun to the Milky Way and the galaxies beyond.

Kit for the year: Projector · Star chart / planisphere · Paper, pencils, protractor · Gnomon / shadow stick · Measuring tape · Binoculars · Dobsonian reflector · Eyepiece set · Red torch

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1 Stars are not all the same

Classroom · 80 min

Bring: Projector, Star chart / planisphere

↪ **Stars Differ** *returns here — activity first run in Class 6, session 9 ("The sky night"). Rerun it with this year's depth.*

> Open the year by asking who has actually looked at the night sky for ten unbroken minutes — in a city class, almost nobody. Show a long-exposure sky photo and ask what differs star to star: brightness, colour, and nothing else the eye can get. Constellations must land as human inventions, not physical objects — draw Orion's stars, then reveal their wildly different distances. Star names are the hook for homework: send them out tonight to find one named star and report back. That homework is the course.'

2 Handspans and star charts

Indoor lab · 80 min

Bring: Star chart / planisphere, Paper, pencils, protractor

↪ **Angles on the Sky** *returns here — activity first run in Class 6, session 3 ("A stick tells the time"). Rerun it with this year's depth.*

↪ **Star Charts & Planetarium Apps** *returns here — activity first run in Class 7, session 8 ("Learning to see in the dark"). Rerun it with this year's depth.*

↪ **Observing Logs & Sketching** *returns here — activity first run in Class 7, session 8 ("Learning to see in the dark"). Rerun it with this year's depth.*

> Calibrate every student's own hand: fist ≈ 10°, span ≈ 20°, against a wall chart they measure with a protractor first. Then hand out the planisphere and make them fail at it before you explain it — set the wrong date, find the 'error', fix it. Start the observing log HERE, this week, with a naked-eye entry (moon position against their fist-measured landmark). A log started in week 2 gets kept; a log started before the observing night does not.

3 The wheeling sky and the star that stands still

Classroom · 80 min

Bring: Projector, Star chart / planisphere

↪ **Diurnal Motion** *returns here — activity first run in Class 6, session 9 ("The sky night"). Rerun it with this year's depth.*

↪ **The Pole Star** *returns here — activity first run in Class 6, session 9 ("The sky night"). Rerun it with this year's depth.*

> Play a star-trail time-lapse before any diagram: the sky visibly turns around one point. Every class contains the misconception that the pole star is the brightest star — put it to a vote, then show Dhruva sitting at magnitude 2, forty-eighth in line. The Dhruva–Saptarshi story is not decoration: 'the seven rishis walk around the fixed prince' IS diurnal motion, taught by grandmothers for centuries. Show how the Saptarshi pointer stars find Dhruva; that trick gets tested on the observing night.

4 Wanderers and the road they keep

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor

↪ **Planets: the Wanderers** *returns here — activity first run in Class 6, session 11 ("The wanderers and the family"). Rerun it with this year's depth.*

> 'Planet' must stop meaning 'a ball in a diagram' and start meaning 'a bright star that cheats' — show the same constellation photographed months apart with a planet moved. Seasonal skies answer a question someone will have asked by now: why can't we see Orion in May? Walk the classroom model: lamp Sun in the centre, zodiac posters on the walls, students as Earth — night side faces different walls in different months. The ecliptic then falls out for free: the wanderers keep to the same twelve posters. Do NOT explain retrograde loops if asked — write the question on the board and promise it to class 9. A kept promise there is worth more than a rushed answer here.

5 Noon: measuring the Earth with a stick

Solar session · 80 min

Bring: Gnomon / shadow stick, Measuring tape, Paper, pencils, protractor

↪ **Measuring the Earth** *returns here — activity first run in Class 6, session 4 ("How big is the Earth?"). Rerun it with this year's depth.*

↪ **Zero Shadow Day** *returns here — activity first run in Class 6, session 6 ("The noon shadow project"). Rerun it with this year's depth.*

> The flagship. A vertical stick at local noon gives the sun's altitude; a partner school a few hundred km north or south (arrange this in week 1, not week 9) gives a second angle and the class computes the circumference of the Earth to within a few percent. Let them do the arithmetic — the wrong answers matter. Zero-shadow day is the local payoff: compute your town's date from its latitude, then check whether your latitude even gets one. If it falls in term time, schedule the five-minute stand-in-the-yard moment; a shadowless student is a photograph they keep.

6 Light: the only messenger

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor

Prism or CD-fragment spectroscope build

Spectrum & Colour — White light is a mixture; a prism fans it into the spectrum, red to violet, each colour a different wavelength. Colour is the first physical measurement of starlight.

Rope and slinky waves: change frequency, see wavelength answer

Waves — Ripples on water and sound in air both carry energy as waves, with a wavelength and a pitch. Light is a wave too — its 'pitch' is colour, and motion can stretch or squeeze it.

↪ **Light & Lookback Time** *returns here — activity first run in Class 7, session 1 ("Light is a traveller"). Rerun it with this year's depth.*

> Everything astronomy knows arrives as light, so spend a period on the messenger itself. Light having a speed is the shock: 1.3 seconds from the Moon, 8 minutes from the Sun — so the Sun you see is 8 minutes old, and someone will ask the right next question about the stars; hold that thread, session 11 pays it. A prism or a water-tray rainbow splits sunlight live — they met the rainbow in class 7, but now frame the spectrum as information carried by the light, not just decoration. Waves are the vocabulary for that: wavelength = colour is the one fact to nail down.

7 Bending light, and the eye at night

Indoor lab · 80 min

Bring: Paper, pencils, protractor, Projector

↪ **Lenses, Mirrors & Focal Length** *returns here — activity first run in Class 7, session 3 ("Glass: lenses, mirrors, and the telescope"). Rerun it with this year's depth.*

↪ **The Eye as a Detector** *returns here — activity first run in Class 7, session 8 ("Learning to see in the dark"). Rerun it with this year's depth.*

↪ **Dark Adaptation & Averted Vision** *returns here — activity first run in Class 7, session 8 ("Learning to see in the dark"). Rerun it with this year's depth.*

> Every pair of hands gets a lens: magnifier + sunlit window = an image of the world upside down on paper. They know refraction from class 7; the new idea is that a curve ORGANISES the bending into an image — that is the whole secret of every telescope they will ever meet. Then the instrument they already own: the eye, with its iris-aperture and its slow chemical night mode. Run the lights-off drill — blackout the lab for ten minutes and have them time how long before they can read a dim card. That number (and why one flash of white light resets it) is the discipline the observing night runs on. Introduce the red torch here and make its logic explicit.

8 Binoculars, telescopes, and what Galileo saw

Indoor lab · 80 min

Bring: Binoculars, Dobsonian reflector, Eyepiece set

↪ **Binocular Astronomy** *returns here — activity first run in Class 7, session 3 ("Glass: lenses, mirrors, and the telescope"). Rerun it with this year's depth.*

↪ **First Light: Using a Telescope** *returns here — activity first run in Class 7, session 3 ("Glass: lenses, mirrors, and the telescope"). Rerun it with this year's depth.*

> Daytime handling drill, no sky needed: focus binoculars on a distant signboard, learn to brace elbows or rest on a wall — hand-shake is the beginner-killer and it is fixable in one afternoon. Then the dobsonian: name the parts, find, centre and focus on a terrestrial target, and rehearse the queue discipline (don't touch the tube, ask for the focuser). End with Galileo's 1610 notebook pages: four dots that moved night after night, and what that did to the idea that everything orbits Earth. The class will re-perform that observation with their own eyes on the observing night if Jupiter is up — check whatsup.html before promising.

9 Evening under the crescent

Observing night · 80 min

Bring: Binoculars, Dobsonian reflector, Eyepiece set, Star chart / planisphere, Red torch

> ### Count stars in the Little Dipper / around Orion to grade your sky
> **Light Pollution** — Artificial skyglow erases faint stars; the Bortle scale grades sky darkness from 1 (pristine) to 9 (inner city). Shielded, warm, dimmed lighting brings stars back.

↪ **Lunar Observing** *returns here — activity first run in Class 7, session 9 ("Night at the eyepiece"). Rerun it with this year's depth.*

↪ **Earthshine** *returns here — activity first run in Class 7, session 9 ("Night at the eyepiece"). Rerun it with this year's depth.*

↪ **The Milky Way Band** *returns here — activity first run in Class 7, session 9 ("Night at the eyepiece"). Rerun it with this year's depth.*

> The night the year points at. A crescent is the right moon three times over: the terminator throws crater shadows a full moon flattens, earthshine is visible ('the old moon in the new moon's arms' — make them explain whose light that is, it is the best viva question of the year), and it sets early enough to leave real darkness. Run the moon first through the dobsonian while twilight fades and eyes adapt. Then charts out, red torches only: find Dhruva by the Saptarshi pointers, log two constellations. If the site is dark enough for the Milky Way band, end there; if it is not, THAT is the light-pollution lesson, taught by the sky itself — count stars in the Saptarshi bowl and compare with a dark-site photo. Every student leaves with a log entry written at the eyepiece, not at home from memory.

10 The Sun is a star

Classroom · 40 min

Bring: Projector

↪ **The Sun is a Star** *returns here — activity first run in Class 7, session 4 ("Our star, up close and safely"). Rerun it with this year's depth.*

> One idea, one period, and it is the pivot of the whole course: the Sun is a star seen close up, and every star is a sun seen from very far away. Run the thought experiment in both directions — drag the Sun away until it is a dot (how far until it looks like Sirius?), drag Sirius in until it is a disc. If the class believes this one sentence, the remaining two sessions are consequences; if they don't, the galaxy is just vocabulary. Take the full forty minutes and take objections seriously.

11 Light years: putting a ruler on the stars

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor, Measuring tape

> ### If the Sun were in Bhopal, where is Alpha Centauri?
> **Light Years & Cosmic Scale** — The nearest star is 4.2 light years away — 7,000× farther than Neptune. Grasping the jump from solar-system scale to interstellar scale is a rite of passage.

> Cash in session 6's IOU: light from Alpha Centauri left 4.3 years ago, so a light year is a RULER, not a duration — the most common class 8 error is filing it under time, and the name is to blame; say so. Scale model in the corridor: Sun a marble at one end, Earth 2.5 cm away, and the nearest star… 700 km away, another town. Let them compute that last number themselves; disbelief is the correct reaction. Then what the darkness between holds: run the Pleiades (which they can find with binoculars) and the Orion Nebula as the two live examples — star cluster and star nursery, both logged targets for their own next dark evening.

12 Our galaxy, and the others

Classroom · 80 min

Bring: Projector

Map naked-eye band features onto a galaxy diagram

Structure of the Milky Way — A barred spiral of 100–400 billion stars, 100,000 ly across, with us in a quiet arm 27,000 ly out. The naked-eye band is our inside view of the disc.

> Close the loop with the observing night: that faint band was our own galaxy viewed from inside — a disc seen edge-on from a seat two-thirds of the way out. The hard cognitive step is being inside the thing being described; the classroom-as-disc walkthrough (look along the rows = many heads = the band, look up = few) does more than any artist's impression, every one of which should be labelled 'not a photograph — nobody has ever left to take one'. Then the kicker: the Andromeda smudge, visible to their naked eye on a dark night, is another entire Milky Way — the farthest thing an unaided human eye can see, and light 2.5 million years old landing on their retina. End the year there, with the size of everything, and hand the observing logs back with a comment each.

CLASS 9

Sky & Space: Class 9

A year of measurement and argument for class 9: coordinates, the evidence for heliocentrism, Kepler to gravitation to ISRO's orbits, the Sun as a star, a hand-plotted H-R diagram, the expanding universe — and the Indic tradition read with the same toolkit.

Kit for the year: Projector · Paper, pencils, protractor · Star chart / planisphere · Measuring tape · Refractor · Solar filter (front aperture) · Tripod · Binoculars · Dobsonian reflector · Eyepiece set · Red torch · Smartphone · Phone-to-eyepiece adapter · Gnomon / shadow stick

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4 Kepler's laws, then Newton's why

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor, Measuring tape

Draw an ellipse with two pins and a loop of string

Kepler's Laws — Orbits are ellipses with the Sun at a focus; planets sweep equal areas in equal times; period squared tracks distance cubed. Three rules that turned astronomy quantitative.

> Order matters more here than anywhere in the course: Kepler FIRST, as pure observed pattern — ellipses, equal areas, $T^2 \propto a^3$ — mined from Tycho's data with no idea why. Have them verify the third law themselves with a table of real planet periods and distances; the moment T^2/a^3 comes out constant eight times running is the moment they believe laws can hide in numbers. Draw ellipses with two pins and a loop of string, always. THEN Newton, as the answer to 'why these patterns?' — one inverse-square attraction explains all three at a stroke, plus the falling apple, plus the Moon. They are doing gravitation in physics class this very term; your job is the part physics class skips — that the law was worth finding because Kepler had found something needing explanation.

5 Orbits: falling forever, and India's rockets

Classroom · 80 min

Bring: Projector

Balloon on a string across the room, then a two-stage balloon: feel why staging wins

Rockets & Launch Vehicles — A rocket throws mass backwards to be pushed forwards, and sheds stages as they empty so it never hauls dead tanks to orbit. India's ladder runs SLV-3 (Rohini, 1980, the first Indian orbital launch), ASLV, the PSLV workhorse, GSLV with its indigenous cryogenic upper stage, LVM3 which sent up Chandrayaan-3, and the small-satellite SSLV.

↪ **India in Space** *returns here — activity first run in Class 6, session 12 (“India in space”). Rerun it with this year's depth.*

> Newton's cannonball kills the year's stubbornest misconception — that astronauts float because there is no gravity up there. Fire the cannon faster and faster until the fall misses the Earth forever; gravity up there is nearly full strength, and an orbit IS falling. Insist on the phrase 'falling around the Earth' until the class uses it unprompted. Then the payoff nobody expects to be local: ISRO flies on exactly this physics. Aryabhata to Chandrayaan-3's south-pole landing to Aditya-L1 staring at the Sun; PSLV as the reliable workhorse and why a rocket is Newton's third law with the reaction mass thrown downward. Launch eastward from Sriharikota and the Earth's spin is free velocity — let them work out why east. Careers land harder here than in any assembly talk.

6 The Sun through a filtered telescope

Solar session · 80 min

Bring: Refractor, Solar filter (front aperture), Tripod, Paper, pencils, protractor

↪ **Refractors & Reflectors** *returns here — activity first run in Class 7, session 3 ("Glass: lenses, mirrors, and the telescope"). Rerun it with this year's depth.*

↪ **Solar Observing (Safely)** *returns here — activity first run in Class 7, session 4 ("Our star, up close and safely"). Rerun it with this year's depth.*

↪ **Sunspots & Solar Activity** *returns here — activity first run in Class 7, session 4 ("Our star, up close and safely"). Rerun it with this year's depth.*

> Safety is the lesson, not the preamble: NOTHING about this session happens until every student has repeated the rule — no eye to any unfiltered optic pointed sunward, ever; a telescope is a magnifying glass and their retina has no pain nerves. Check the front-aperture filter for pinholes in front of the class, deliberately, so they see checking is normal. Teach the designs comparison on the hardware itself first (refractor bends, reflector bounces — open the dob's tube end and show the mirror) while pairs sketch. Then the Sun: sunspots as magnetic storms larger than Earth — draw today's spots, and if you can manage a second look days later, the rotation of the Sun falls out of two sketches. Log it like any observation; the Sun is the one star they will ever see as a disc, which is precisely what sessions 7–10 exploit.

7 How bright, how far: magnitudes and parallax

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor

> ### Thumb-blink parallax demo, then scale to real star shifts
> **Stellar Parallax** — Nearby stars shift minutely against the background as Earth orbits — the parallax that defines the parsec and anchors every distance in astronomy.

↪ **The Inverse-Square Law** *returns here — activity first run in Class 7, session 2 ("The Sun is an engine"). Rerun it with this year's depth.*

> The magnitude scale is historical debris — backwards (bigger = fainter) and logarithmic — so teach it as Hipparchus's fault and move on; five steps = 100×, Vega ≈ 0, faintest naked eye ≈ 6, done. Inverse-square is the physics under everything stellar: same torch, twice the distance, quarter the light — do it with a phone torch and a lux-meter app on the lab bench, four data points. Then the year's most elegant measurement: hold up a thumb, blink alternate eyes, watch it jump against the far wall. Baseline = eye spacing. Now let the baseline be the Earth's orbit and the thumb be a star: parallax, the ONLY rung of the distance ladder that is pure geometry, no assumptions. Be honest that the angles are fractions of an arcsecond — which is why the light year had to wait for telescopes, and why class 8's Alpha Centauri number was hard-won.

8 Starlight decoded: colour, temperature, spectra

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor

Rank Betelgeuse, Capella, Sirius, Rigel by temperature by eye

Star Colours & Temperature — A star's colour is a thermometer reading: red ~3,000 K, yellow ~6,000 K, blue-white 10,000 K+. Hotter objects glow bluer — physics you can check with your own eyes.

View street-lamp and CFL spectra through a diffraction grating

Spectroscopy — Elements imprint fingerprint lines on a spectrum. From those lines alone we read a star's composition, temperature and motion — how we know what stars are made of.

> Start by breaking the childhood colour code: in a flame or a filament, BLUE is hotter than red — the iron-in-the-forge sequence runs dull red, orange, yellow, white, blue-white. So Betelgeuse (red) is cool and Rigel (blue) is fierce, and star colour is a thermometer read across light years. Then the deeper trick: a slit of light through a grating (a CD sliver taped to card works; build one per pair from stationery) shows lines — barcodes, one pattern per element, the same in the lab and in a star. Someone should gasp at the implication and if nobody does, say it plainly: we know what stars are MADE OF without going. Kill the 19th-century philosopher Comte's famous 'we shall never know the composition of stars' as the best wrong prediction in science. Both threads — temperature and identity — braid into session 9's diagram.

9 The H-R diagram, plotted by hand

Indoor lab · 80 min

Bring: Paper, pencils, protractor, Projector

Torch at 1 m vs 10 m: inverse-square in the classroom

Luminosity & Absolute Magnitude — Apparent brightness mixes true output with distance; absolute magnitude removes distance (all stars imagined at 32.6 ly). Only then can stars be honestly compared.

Plot 30 bright stars from catalogue data

The H-R Diagram — Plot luminosity against temperature and stars fall into families: the main sequence, giants, white dwarfs. One scatter plot that encodes how stars live and die.

> First close the brightness loophole: apparent magnitude confounds 'truly bright' with 'merely close', so park every star at the same standard distance (10 parsecs — via session 7's parallax and inverse-square) and compare honestly; that is absolute magnitude, and the Sun's humiliating drop from −27 to +4.8 is the number to dwell on. Then the year's centrepiece, and it must be done by HAND: give pairs a table of ~30 real stars (colour/temperature vs absolute magnitude), graph paper, forty minutes. Do not pre-draw the axes' meaning — let them discover the diagonal band themselves. The main sequence emerging out of their own plotted dots is the closest a classroom gets to the 1911 moment: stars are not scattered at random, and a scatter plot just became the map of stellar physics. Name the corners (giants, dwarfs) only after every pair has found the band.

10 The lives of stars

Classroom · 80 min

Bring: Projector

Infrared vs visible views of the same nursery

Star Birth — Cold gas clouds collapse under gravity, heat up, and ignite fusion. The Orion Nebula is a stellar nursery in action, visible in binoculars.

Sort star cards into an evolutionary sequence by mass

Lives of Stars — Stars balance gravity against fusion for millions to trillions of years, then swell to giants and die — gently as white dwarfs or violently as supernovae. Mass decides everything.

> Yesterday's map becomes today's story. Birth first: a cold cloud like class 8's Orion Nebula, gravity (session 4's, same law) pulling it together until the core lights — the main sequence band they plotted is not a road stars travel but where stars SIT while fusing hydrogen, the single most common misconception at this level; attack it explicitly with the 'photograph of a crowd' analogy — babies, adults, elders all in one frame, and the H-R diagram is that photograph. Mass is destiny: heavier stars burn brighter and die younger, profligately. Walk the Sun's own future — swelling to a red giant (the class 8 'Sun is a star' pivot now runs in reverse), shedding, ending as a white dwarf — with the honest timescale of five billion years, which is also the reassurance. End at the fact that makes teenagers sit up: the calcium in their bones was cooked inside a star that died before the Sun was born.

11 The expanding universe

Classroom · 80 min

Bring: Projector, Paper, pencils, protractor

Ambulance-siren analogy + real galaxy spectra

Doppler & Redshift — Motion stretches or squeezes light waves: approaching sources shift blue, receding ones red. Measured in spectral lines, it gives speeds of stars and galaxies to high precision.

Inflate a dotted balloon: every dot recedes from every other

The Expanding Universe — Nearly every galaxy's light is redshifted, and farther means faster — space itself is stretching. Run the film backward and the universe had a beginning.

Cosmic history on a 1-year calendar (Sagan style)

Big Bang & Cosmic History — 13.8 billion years from a hot dense start: relic microwave glow, primordial hydrogen and helium, and the growth of galaxies all tell one consistent story.

> Three ideas, one crescendo, and each stands on something the class already owns. Doppler: they have HEARD it (every passing bike horn drops in pitch); light does it too, and session 8's spectral barcodes are the ruler — lines shifted redward mean recession, and the shift is a speedometer. Hubble: nearly every galaxy's barcode is redshifted, and the farther, the faster. Run the balloon with sticker-galaxies and let every sticker be 'home' — everyone sees everyone else receding, no centre, which defuses 'where did it bang?' before anyone asks. Then run the film backwards to a hot dense beginning ~13.8 billion years ago. Be scrupulous about what Big Bang cosmology is — an extrapolation from measured expansion, model-thinking at full stretch — and end the physics year on the scale joke the course has earned: class 8 ended at Andromeda; class 9 ends at everything.

12 Observing night: star-hopping the first-quarter sky

Observing night · 80 min

Bring: Binoculars, Dobsonian reflector, Eyepiece set, Star chart / planisphere, Red torch

Hop from Vega to M57 or Saptarshi to M51

Star Hopping — Navigating from a bright star to a faint target through chart-matched patterns, hop by hop. The craft skill that turns a chart reader into an observer.

> One skill tonight, practised until it is theirs: navigating chart-to-sky in hops — anchor on a bright star, walk a pattern of steps each smaller than the binocular field, arrive at something invisible to the naked eye. Choose two hop routes for the season in advance and dry-run them yourself the night before; a teacher lost mid-hop teaches the wrong lesson. This is the year's coordinates made physical — the chart is in equatorial coordinates and their fist is still 10°. The first-quarter moon is deliberate, not a compromise: craters along the terminator are at their best through the dob for the queue, and the moon sets by midnight. Students who finish a hop early get the meridian question: which of tonight's targets is transiting right now, and how do you know? Log everything at the eyepiece.

13 Imaging night: the moon on their phones

Imaging night · 80 min

Bring: Dobsonian reflector, Eyepiece set, Smartphone, Phone-to-eyepiece adapter, Red torch

↪ **Smartphone Astrophotography** *returns here — activity first run in Class 7, session 9 (“Night at the eyepiece”). Rerun it with this year's depth.*

> The very next evening, same moon, one goal: every student leaves with a moon photograph THEY took, on THEIR phone. The terminator craters are the one astrophotography target that succeeds on the first attempt, which is why this night exists — success breeds astronomers, and a phone photo travels to the whole family by bedtime, which is your outreach programme running itself. Technique is three rules: adapter (or two-handed brace) against the eyepiece, tap-and-hold to lock focus and drag exposure DOWN — every phone will try to brighten the moon into a white blob — and let a timer or volume-button shutter fire the shot, never a screen jab. Have them shoot ten and keep one; culling is the real lesson. Print the best three for the classroom wall with the photographers' names.

14 Nakshatra, rashi, and the astrology question

Classroom · 80 min

Bring: Projector, Star chart / planisphere

Design a fair test for an astrological claim; discuss what results would mean

Astronomy vs Astrology — Same sky, different claims: astronomy predicts where planets are; astrology claims what they mean. One invites testing, the other resists it — a respectful, evidence-first discussion every Indian classroom deserves.

> Teach the Indic sky as what it technically is — coordinate systems centuries old. The 27 nakshatras divide the moon's monthly path (13°20' each, one per night of the sidereal month: the moon 'staying in a different lodge' nightly is an OBSERVATION their grandparents' calendar encodes); the 12 rashis divide the Sun's yearly path — the same ecliptic from class 8, sliced two ways for two clocks. Find the class's janma nakshatras on the chart; engagement is instant and personal. Then the question every class 9 room is silently asking, treated with respect and the year's own toolkit: astrology is a MODEL, so test it the way session 3 tested geocentrism — what does it predict, and do the predictions hold when checked blind? Present the evidence and let the model fail on its own terms. Never mock — half the room's families consult panchangs — but be exact: the failure of astrology as prediction and the brilliance of the astronomy underneath it are BOTH true, and the second is the inheritance.

15 Jantar Mantar: the observatory with no lens

Indoor lab · 80 min

Bring: Gnomon / shadow stick, Measuring tape, Paper, pencils, protractor, Projector

Build a shanku and derive latitude from noon shadow

Traditional Instruments — Shanku (gnomon), ghati yantra (water clock), chakra yantra (graduated ring) and the astrolabe: pre-telescopic instruments that measured time and angle to surprising precision.

Field trip / virtual tour: read time off the Samrat Yantra

Jantar Mantar Observatories — Jai Singh II's giant masonry instruments (1720s) — the Samrat Yantra sundial reads time to 2 seconds. Architecture-scale astronomy you can still walk through in Jaipur and Delhi.

> End the year in the school ground, closing the oldest loop in the course. Start from the humble gnomon they have used since class 6, and scale it in imagination: what if the stick were 27 metres tall? The shadow edge moves a measurable millimetre every few seconds — sheer SIZE buys precision, which is the entire engineering idea of Jai Singh's 1720s observatories. Tour the instruments as solved problems, not monuments: the Samrat Yantra is a sundial big enough to read to two seconds; the meridian instruments do exactly what session 2 promised a transit-timing wall could do. Then the closing exercise, pairs, on paper: design a naked-eye instrument to measure one thing — a star's altitude, local noon, the Sun's solstice extremes — using only masonry and geometry. Judge the designs against Jai Singh's. Some student's wall-with-a-slit will essentially reinvent one of his instruments, and that is the note to end the year on: the toolkit they now carry was always the whole game.